

Q1 B: Given a square matrix A , prove analytically that the trace of A is equal to the sum of its eigenvalues. Provide a numerical example to illustrate this property using a specific matrix.

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1 Definitions

Definition 1 (Trace). The **trace** of an $n \times n$ square matrix $A = (a_{ij})$ is the sum of its main diagonal entries:

$$\text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

Definition 2 (Eigenvalue). A scalar $\lambda \in \mathbb{C}$ is an **eigenvalue** of A if there exists a non-zero vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. The eigenvalues are exactly the roots of the **characteristic polynomial**

$$p(\lambda) = \det(\lambda I - A).$$

2 The Characteristic Polynomial and Vieta's Formulas

Lemma 3 (Structure of the Characteristic Polynomial). *For an $n \times n$ matrix A with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (counted with algebraic multiplicity), the characteristic polynomial is*

$$p(\lambda) = \det(\lambda I - A) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n) = \lambda^n - \left(\sum_{i=1}^n \lambda_i\right) \lambda^{n-1} + \cdots + (-1)^n \det(A).$$

In particular, the coefficient of λ^{n-1} is $-\sum_{i=1}^n \lambda_i$.

Proof. Since the eigenvalues are precisely the roots of $p(\lambda)$ and p is monic of degree n , the factored form follows from the Fundamental Theorem of Algebra. Expanding the product $(\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n)$, the coefficient of λ^{n-1} is $-(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$, which is Vieta's formula for the sum of the roots. $\square$

3 Main Theorem: $\text{tr}(A) = \sum \lambda_i$

Theorem 4. Let A be an $n \times n$ matrix over $\mathbb{C}$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (counted with algebraic multiplicity). Then

$$\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n.$$

Proof. We expand $p(\lambda) = \det(\lambda I - A)$ directly and compare the coefficient of λ^{n-1} with that obtained from the factored form in Lemma 3.

Step 1: Expand $\det(\lambda I - A)$.

Write

$$\lambda I - A = \begin{pmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{pmatrix}.$$

The determinant is a sum over all permutations $\sigma \in S_n$:

$$\det(\lambda I - A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n (\lambda I - A)_{i, \sigma(i)}.$$

Step 2: Identify which terms contribute to λ^{n-1} .

Each factor $(\lambda I - A)_{i, \sigma(i)}$ equals:

- $(\lambda - a_{ii})$ if $\sigma(i) = i$ (diagonal entry), contributing a λ or a constant;
- $-a_{i, \sigma(i)}$ if $\sigma(i) \neq i$ (off-diagonal entry), a constant.

A term $\prod_{i=1}^n (\lambda I - A)_{i, \sigma(i)}$ has degree k in λ , where k is the number of fixed points of σ (positions where $\sigma(i) = i$).

To obtain λ^{n-1} , we need *exactly* $n-1$ fixed points. But a permutation with $n-1$ fixed points must fix all n points (since fixing $n-1$ forces the last to be fixed too). Therefore the *only* permutation contributing to λ^{n-1} is the **identity** $\sigma = \text{id}$.

Step 3: Compute the contribution of the identity permutation.

For $\sigma = \text{id}$, $\text{sgn}(\text{id}) = 1$ and

$$\prod_{i=1}^n (\lambda - a_{ii}) = \lambda^n - (a_{11} + a_{22} + \dots + a_{nn})\lambda^{n-1} + \dots = \lambda^n - \text{tr}(A)\lambda^{n-1} + \dots$$

All other permutations $\sigma \neq \text{id}$ have at most $n-2$ fixed points, so they contribute terms of degree at most $n-2$ in λ .

Step 4: Equate coefficients.

From Steps 2–3, the coefficient of λ^{n-1} in $p(\lambda) = \det(\lambda I - A)$ is $-\text{tr}(A)$.

From Lemma 3, the coefficient of λ^{n-1} in $(\lambda - \lambda_1) \cdots (\lambda - \lambda_n)$ is $-\sum_{i=1}^n \lambda_i$.

Therefore:

$$-\text{tr}(A) = -\sum_{i=1}^n \lambda_i \implies \text{tr}(A) = \sum_{i=1}^n \lambda_i. \quad \square$$

Remark. The same argument shows that $(-1)^n \det(A)$ is the constant term of $p(\lambda)$, i.e. $\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$.

4 Explicit Verification for 2×2 and 3×3 Matrices

4.1 The 2×2 Case

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$\det(\lambda I - A) = (\lambda - a)(\lambda - d) - bc = \lambda^2 - (a + d)\lambda + (ad - bc).$$

By Vieta's formulas: $\lambda_1 + \lambda_2 = a + d = \text{tr}(A)$. ✓

4.2 The 3×3 Case

Let $A = (a_{ij})_{3 \times 3}$. The diagonal product gives

$$(\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33}) = \lambda^3 - (a_{11} + a_{22} + a_{33})\lambda^2 + \dots$$

Every other term in $\det(\lambda I - A)$ involves at most one diagonal factor, contributing at most λ^1 . Hence the λ^2 coefficient is exactly $-(a_{11} + a_{22} + a_{33}) = -\text{tr}(A)$, confirming $\lambda_1 + \lambda_2 + \lambda_3 = \text{tr}(A)$. ✓

5 Numerical Examples

5.1 Example 1: 2×2 Matrix

Example 5. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Trace:

$$\text{tr}(A) = 1 + 4 = 5.$$

Characteristic polynomial:

$$p(\lambda) = \det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 4 \end{vmatrix} = (\lambda - 1)(\lambda - 4) - 6 = \lambda^2 - 5\lambda - 2.$$

Eigenvalues (quadratic formula):

$$\lambda = \frac{5 \pm \sqrt{25 + 8}}{2} = \frac{5 \pm \sqrt{33}}{2}.$$

Sum of eigenvalues (Vieta's formula directly from the polynomial):

$$\lambda_1 + \lambda_2 = 5 = \text{tr}(A). \quad \checkmark$$

5.2 Example 2: 3×3 Matrix

Example 6. Let

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

(Upper triangular matrix — eigenvalues are the diagonal entries.)

Trace:

$$\operatorname{tr}(A) = 2 + 3 + 5 = 10.$$

Characteristic polynomial: Since A is upper triangular,

$$p(\lambda) = (\lambda - 2)(\lambda - 3)(\lambda - 5).$$

Eigenvalues:

$$\lambda_1 = 2, \quad \lambda_2 = 3, \quad \lambda_3 = 5.$$

Sum of eigenvalues:

$$\lambda_1 + \lambda_2 + \lambda_3 = 2 + 3 + 5 = 10 = \operatorname{tr}(A). \quad \checkmark$$

5.3 Example 3: 3×3 Matrix with Non-Trivial Eigenvalues

Example 7. Let

$$A = \begin{pmatrix} 4 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Trace:

$$\operatorname{tr}(A) = 4 + 3 + 2 = 9.$$

Characteristic polynomial: Upper triangular, so

$$p(\lambda) = (\lambda - 4)(\lambda - 3)(\lambda - 2).$$

Eigenvalues:

$$\lambda_1 = 4, \quad \lambda_2 = 3, \quad \lambda_3 = 2.$$

Sum of eigenvalues:

$$4 + 3 + 2 = 9 = \operatorname{tr}(A). \quad \checkmark$$

6 Summary

Matrix	Size	$\operatorname{tr}(A)$	Eigenvalues	$\sum \lambda_i$
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	2×2	5	$\frac{5 \pm \sqrt{33}}{2}$	5
$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 5 \end{pmatrix}$	3×3	10	2, 3, 5	10
$\begin{pmatrix} 4 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \end{pmatrix}$	3×3	9	4, 3, 2	9

Conclusion. The proof proceeds in two stages:

1. Expand $\det(\lambda I - A)$ and show the coefficient of λ^{n-1} is $-\operatorname{tr}(A)$ (only the identity permutation contributes).
2. Factor $\det(\lambda I - A) = \prod(\lambda - \lambda_i)$ and apply Vieta's formula: the coefficient of λ^{n-1} is $-\sum \lambda_i$.

Equating the two expressions gives $\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$.